Clinton Enwerem

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Summary

Robotics Research Engineer with 5+ years of research experience and Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026. Develop demonstration pipelines, evaluate imitation/RL policies, and implement Python/C++ sensing and control software for dexterous manipulation in simulation and on physical robots.

Skills

Robotics: cuRobo, ROS/ROS 2, ros2_control, MoveIt 2, MoveIt Servo, SocketCAN, VR teleoperation; **Planning & Control:** RRT*, FCL, state estimation, CBF-CLF QPs, MPC; **Learning:** PyTorch, OpenVLA-7B, QLoRA, ACT, DAgger, Diffusion Policy, PPO, flow matching; **Simulation:** Isaac Lab/Sim, Gazebo, MuJoCo, Drake; **Software:** Python, C++, Linux, Docker, git, CI; **GPU Computing:** CUDA, Warp, MuJoCo Warp, JAX; **Perception:** FoundationPose, BundleSDF, SAM 2, SAM 3D, PCL, OpenCV; **Robots:** Fairino FR3, xArm7, UR3e, UR5, Robotiq Hand-E, RealHand A7 (Bimanual Arm-Hand System), RealHand L6, LEAP Hand v1, Unitree G1 (simulation).

Work Experience

Institute for Systems Research (ISR), University of Maryland Aug. 2021 – Present
Graduate Research Assistant *College Park, MD*
• Develop grasp-synthesis, state-estimation, motion-planning, and feedback-control methods for dexterous manipulation under contact uncertainty; implement Python/C++ and ROS 2 software for physical arm–hand systems.
• Trained PPO with GDF rewards in **1,024 Isaac Lab environments**: **64/64** pregrasp arrivals versus **0/64** with palm-distance rewards.
• Developed xArm7-RealHand-L6 VR teleoperation with wrist-to-Cartesian velocity mapping, relative joint-position control, grasp primitives, thumb abduction, and hand reset; integrated MoveIt Servo with Dimensional OS (DimOS) for collision detection, arm speed modulation, and velocity damping near kinematic boundaries.
• Fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing; each of three training seeds achieved 136/200 task completions versus 129/200 without supervision, recovering seven failures without disrupting nominal successes.
• Developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. Evaluated task execution in MuJoCo.

MATRIX Lab, USM at Southern Maryland Jun. – Aug. 2022
Research Intern *California, MD*
• Identified a discrete-time Crazyflie model from flight data and implemented a finite-horizon optimal formation controller in Python and Crazyswarm; executed leader–follower formation flights with 10 physical Crazyflie quadrotors under localization uncertainty.

Kognitive Robotics Mar. 2021 – Aug. 2021
Robotics Engineer *Lagos, Nigeria*
• Designed power and sensor–actuator interfaces in Altium; assembled a mobile robot and developed and tested ROS software for the platform.

Robotics & Artificial Intelligence Nigeria (RAIN) Jan. 2020 – Feb. 2021
Robotics Trainee *Ibadan, Nigeria*
• Developed a ROS-based quadrotor autonomy stack on a Raspberry Pi 3, integrating ORB-SLAM3 visual SLAM, an EKF fusing SLAM poses with IMU data, and a PID flight controller. The onboard controller maintained stable hover during ground-station packet loss.

Projects

ParcelStow — Execution-Timing Robustness in Imitation Learning (Paper, project, Code, Data & Models) | *Isaac Lab, PyTorch* 2026
• Built ParcelStow, an Isaac Lab benchmark for assessing imitation-policy robustness to execution-timing variation; generated **970K** expert states (**297** demonstrations) and trained ACT, Diffusion Policy, and DAgger. Controlled timing stress tests exposed precision-insertion failures: at 2× execution speed, ACT achieved **53%** success versus **84%** for the expert.

Grasp Distance Fields — Feedback Grasp Execution (Project, Paper) | *C++, Python, QP* 2026
• Implemented a CBF-CLF quadratic-program controller for feedback grasp execution; simulated experiments lifted **46 of 50** objects, with a median QP solve time of **0.09 ms**.

dexreach — GPU-Accelerated Dexterous Grasp and Motion Planning (Project) | *cuRobo, PyTorch, Isaac Lab* 2026
• Built GPU-accelerated grasp synthesis and cuRobo motion planning with friction/actuator checks and Isaac Lab validation: dual Franka-Ability completed both objects' sequences in **6/6** tested plans, with **4–19 mm** final position errors.

Selected Publications

• **C. Enwerem et al.** "Does Imitation Learning Preserve Temporal Robustness in Dexterous Manipulation? An Expert–Learner Comparison Across Task Execution Speeds." *arXiv preprint*, 2026. arXiv:2609.01453, project page, code, data & models.
• **C. Enwerem et al.** "Grasp Execution Without a Planner: Configuration-Space Grasp Distance Fields with Certified Safety & Guaranteed Quality." *Preprint*, 2026. arXiv:2608.00600, project page.
• **C. Enwerem et al.** "Safety-Aware Reinforcement Learning for Control via Risk-Sensitive Action-Value Iteration and Quantile Regression." *CDC 2025*. arXiv:2506.06954.
• Additional first-author papers in control, safe learning, and robotic manipulation: IROS 2026; CDC 2026/2025/2024. Full list: Google Scholar.

Education

Ph.D., Electrical & Computer Engineering, University of Maryland, College Park, MD. Expected Fall 2026. Aug. 2021 – Present
B.Eng., Electrical Engineering, University of Nigeria, Nsukka. Highest Honors. GPA 3.84/4.0. Aug. 2018

Clinton Enwerem

Santa Clara, CA, United States

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Dear Technical Recruiter,

Application: Robotics Research Engineer

I implement Python/C++ robotics software, including ROS 2 hardware interfaces, robot descriptions, controllers, motion planners, simulations, and automated tests, then validate the software on physical robots. I am an Electrical and Computer Engineering Ph.D. candidate at the University of Maryland, graduating in Fall 2026, with five years of work in dexterous manipulation, motion planning, feedback control, and robot learning.

realhand_l6_ros2 implements a C++ SocketCAN hardware interface and exports joint and tactile state. The package includes contact-gated control, URDF/SRDF, MoveIt 2, mock hardware, virtual-CAN tests, CI, and physical communication and sensing tests. kinematic_planner_ros2 implements RRT*, Informed RRT*, and FCL primitive, mesh, and self-collision queries for N-DOF URDF manipulators. Grasp Distance Fields enforce collision and grasp constraints with a CBF-CLF quadratic program, which solved in 0.09 ms in the reported manipulation experiments.

ParcelStow evaluates imitation policies in simulation. I trained DAgger, Diffusion Policy, and ACT on 297 expert demonstrations of a Unitree G1 and RealHand L6 insertion task. At twice nominal execution speed, expert success was 84% and ACT success was 53%, with 35 of ACT's 47 failures occurring during insertion. The public release contains the repository, data and models, and stage-level outcomes required to reproduce the comparison.

I fine-tuned OpenVLA-7B with QLoRA for recovery supervision in simulated bimanual packing. Each of three training seeds completed 136/200 tasks versus 129/200 without supervision, recovering seven failed executions without disrupting nominal successes. I developed a visuo-tactile bimanual assembly pipeline with simulation and hardware execution interfaces, combining depth-based pose estimation (RANSAC/DBSCAN, ICP), collision-constrained trajectory optimization (CasADi/IPOPT), and gravity-compensated position control with fingertip-contact monitoring. I evaluated task execution in MuJoCo.

I developed dexreach to connect GPU-accelerated dexterous grasp synthesis and cuRobo motion planning with Isaac Lab sequence validation, evaluating object motion, task-stage completion, and contact-loss stops.

My resume is attached. You can reach me at **enwerem@umd.edu** or **301-416-9430**.

Thank you for considering my application. I am available to start in December 2026. I look forward to discussing how my experience in robot integration, motion planning, and control could support your team's work.

Sincerely,

Clinton Enwerem

October 27, 2026

Department of Electrical & Computer Engineering University of Maryland, College Park, MD 20742

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